

Utilization of agricultural waste-based fillers from rice husk ash and bagasse ash in asphalt concrete-binder course (AC-BC) pavement

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Abstract. The increasing emphasis on sustainable infrastructure has prompted the utilization of alternative and eco-friendly materials in pavement construction. This study investigates the potential use of rice husk ash (RHA) and Bagasse ash (BA), both agricultural waste by-products, as alternative fillers in Asphalt Concrete–Binder Course (AC-BC) mixtures. A quantitative experimental approach was used, involving laboratory tests on aggregates, asphalt, and fillers, followed by Marshall testing. Filler contents of 0%, 2%, 4%, 6%, and 8% were evaluated for stability, flow, void in mix (VIM), voids in mineral aggregate (VMA), voids filled with bitumen (VFB), and Marshall Quotient (MQ). The optimum asphalt content was 6.65%. Results showed that BA improved the mixtures mechanical properties, especially at 4% filler, achieves the highest stability and MQ. RHA performed well at lower levels, but excessive use increased stiffness and risk of cracking. Both fillers meet Indonesian National Standard (SNI) specifications, proving their technical feasibility. Furthermore, their application contributes to waste reduction and environmental sustainability by recycling biomass waste into valuable construction materials. This research confirms that RHA and BA are promising eco-friendly alternatives to conventional mineral fillers, aligning with sustainable infrastructure goals without compromising the structural integrity of pavement layers.

1 Introduction

Road pavements in tropical climates are vulnerable to early failure because of high surface temperatures, frequent heavy rainfall, and substantial traffic loading. Previous studies have shown that elevated temperatures combined with heavy loads to intensify rutting, while moisture and rainfall accelerate the decline in pavement performance and service life [1].

Asphalt mixtures require reinforcement through additives mixed into the asphalt to modify its properties, preventing brittleness at low temperatures and reducing the risk of cracking [2]. Moreover, modification also aims to introduce new characteristics to asphalt and to improve its stability and strength.

In road pavement mixtures, the materials used typically consist of asphalt, fine aggregates, coarse aggregates, and filler. Fillers are materials used to fill the voids between fine and coarse aggregates and can be derived from natural or manufactured crushed rock. Examples of fillers that can be used include agricultural by products including rice husk ash and bagasse ash [3, 4].

The transition toward sustainable infrastructure has become a global concern, including in Indonesia, especially as flexible pavements are considered more efficient in their application. The commonly used fillers

are Portland cement or stone dust. As environmental awareness increases, along with the growing volume of agricultural and industrial waste, there is now a growing push to find alternative biomass-based fillers that are more environmentally friendly and sustainable.

Moreover, the shift toward sustainable construction materials is also driven by the need to mitigate environmental degradation caused by excessive extraction of natural resources. Agricultural wastes materials, specifically RHA and BA, offer not only environmental benefits but also potential improvements in mechanical performance when used as fillers in asphalt mixes.

Utilizing agricultural waste, such as rice husk ash and bagasse ash, offers a sustainable approach to producing environmentally friendly fillers. Rice husk ash contains a high pozzolanic material, which enhances rutting and moisture resistance, making it suitable for increasing the durability of pavement layers and resisting damage from water [5]. Additionally, Bagasse ash contains significant pozzolanic components, particularly silica (SiO₂) which contribute to improved resistance against sulfate and sulfuric acid attacks, chloride ion ingress, and alkali silica reaction (SAR) [6].

The high silica content and angular morphology of RHA contribute to increased Marshall stability at 6-8% filler levels by enhancing binder interaction and mixture

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stiffness [7]. Its pozzolanic reactivity further strengthens cohesion and resistance to deformation. Similarly, BA improves asphalt stability through its angular surface texture and silica presence [8]. The performance of BA is influenced by particle fineness and carbon content, finely ground, low carbon BA optimized packing density and binder bonding, whereas coarser or carbon-rich BA diminishes mixture uniformity [9].

At 6-8% content, RHA's fine particle size and high silica effectively lower both flow and VFB [10]. Its angular and porous structure facilitates mechanical interlock with aggregates, improving load transfer and minimizing air voids. BA can also reduce void content when processed to a fine size and with minimal residual carbon. Well-prepared BA increases mix density and decreases permeability.

Previous studies have demonstrated the utilization of agricultural waste materials such as Rice Husk Ash and Bagasse Ash (BA) as fillers in pavement mixtures. For RHA, several investigations have reported the use of filler contents around 6.5%, 7.0%, and 7.5% [11]. Meanwhile, BA has generally been applied at levels of 4%, 5.5%, and 6.5% [12] and even up to 8% [13]. Overall, the filler content commonly used in previous research ranges from 0-8%, with optimal performance frequently observed between 4% and 6%. Based on these findings, this study employs filler variations of 0%, 2%, 4%, 6% and 8% to represent both the lower and upper boundaries of the commonly reported range.

One of the commonly used types of pavements is Asphalt Concrete (AC), particularly AC-BC (Asphalt Concrete – Bearing Course). This type of asphalt concrete is more resistant to plastic deformation and less prone to cracking.

Indonesia's large agricultural sector produces a significant amount of waste. Therefore, this research seeks to utilize agricultural waste that was previously considered useless into something valuable. RHA and BA both agricultural waste products, are implemented as filler substances in asphalt pavement formulations in this examination. The primary objective involves assessing the influence of utilizing these agricultural residues as filler components. This application of agricultural waste-based fillers is an effort to support environmentally friendly filler material utilization [14].

2 Materials and methods

This research employs a quantitative methodology involving experimental procedures on multiple test samples to achieve target material compositions via laboratory examinations compliant with Indonesian National Standards. The methodology consists of several main stages, including material testing, mix design without filler, mixture development incorporating Rice Husk Ash (RHA) and sugarcane Bagasse Ash (BA) fillers, and determination of the optimal composition for each mixture variation.

2.1 Material

The materials used include coarse and fine aggregates combined with a filler and mixed with 60/70 penetration grade asphalt. The filler variations using RHA and BA are added at 0%, 2%, 4%, 6%, and 8% of the total weight (passing #200 sieve aggregate). The chemical composition of both ashes plays a pivotal role in determining their behavior in asphalt mixtures. RHA is known for its high SiO₂ content, while BA contains more CaO and MgO, contributing to better binding reactivity [14, 15]. Figure 1 and Figure 2. Illustrate the fillers (RHA and BA) used as particle replacements in the binder.



Fig. 1. Rice husk ash

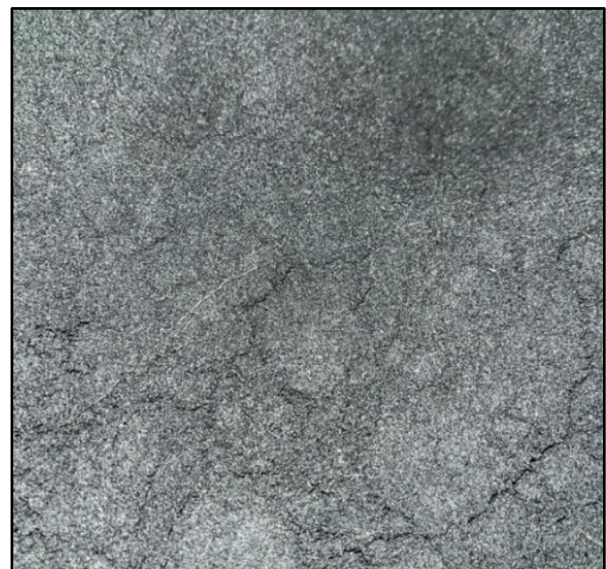


Fig. 2. Bagasse ash

The filler materials investigated in this research, specifically rice husk ash and bagasse ash are depicted in figures 1 and 2 representing the laboratory testing components utilized in the experimental program.

2.2 Mix design

The mixture formulation was established through the Marshall method protocols to identify the Optimum Asphalt Content (OAC). Upon determining the OAC, Marshall testing was conducted by integrating rice husk ash and bagasse ash as filler additives into the composite matrix. Assessment criteria encompassed stability, flow, voids filled with bitumen, Void In mix and mineral aggregate. Aggregate gradation adhesion to asphalt concrete Binder Course requirement specified in Bina Marga specification (2018).

Aggregate gradation adhered to asphalt specified is carried out using a series of Marshall method procedures to determine the Optimum Asphalt Content (OAC). After obtaining the OAC, Marshall tests are performed by adding rice husk ash and sugarcane bagasse ash fillers to the material composition. The parameters analyzed include Stability, Flow, Voids in Mix (VIM), Voids Filled with Bitumen (VFB), and Voids in Mineral Aggregate (VMA). The aggregate proportions are determined according to the Asphalt Concrete - Binder Course (AC-BC) gradation based on the Bina Marga Specification (2018 Revision 2).

2.3 Testing procedure

The testing procedure consists of three main stages, including:

2.3.1 Aggregate and filler testing

Aggregate assessments were performed to evaluate the suitability of aggregates for use in pavement mix materials. The tests are carried out by the procedures and provisions of the 2018 Bina Marga General Specifications, Revision 2. The tests include:

- a. Sieve analysis
- b. Coarse aggregate specific gravity and water absorption
- c. Fine aggregate specific gravity and water absorption
- d. Los Angeles Abrasion Test

2.3.2. Asphalt testing

Asphalt testing is conducted to ensure the asphalt used in the mix meets the specified standards. The tests include:

- a. Penetration test of rubber-modified asphalt
- b. Flash point and fire point test using the Cleveland Open Cup method
- c. Ductility test of rubber-modified asphalt
- d. Specific gravity test of hard asphalt

- e. Softening point test
- f. Asphalt adhesion test
- g. Loss on heating test

2.3.3 Marshall testing

Marshall testing is performed after obtaining the Theoretical Asphalt Content (TAC). The values obtained from the Marshall test are recorded in the Marshall table to determine the Optimum Asphalt Content (OAC). The OAC is determined by varying the asphalt content, plotting the results in typical graphs, identifying the asphalt content that meets the specifications, and calculating the average value from the most frequently compliant data across each typical graph. The standard Marshall test uses 15 specimens with 5 different asphalt content levels. For the durability Marshall test, 12 specimens are used with 4 different soaking durations. Immersion in a 60 °C water bath for 30 minutes consistent with AASHTO T245 standard, simulating typical tropical field conditions and evaluating the mix's resistance to thermal degradation.

3 Result and discussion

Based on the laboratory test results, the physical and technical characteristics of each asphalt concrete mix material namely aggregate, asphalt, and filler were obtained. These tests evaluated the physical and mechanical properties of AC-BC mix materials. The results serve as the basis for evaluating the suitability of each material. The materials tested in the laboratory include aggregates, asphalt, and filler.

3.1 Aggregates and filler

The test results for aggregates and filler were obtained from laboratory testing procedures conducted by Indonesian National Standards (SNI). A summary of the physical characteristics of the aggregates is presented in the following Table 1.

Based on Table 1, it can be observed that all aggregate testing parameters meet the applicable Indonesian National Standards. The specific gravity and absorption values of the aggregates indicate suitable density and characteristics for use in flexible pavement structures. In addition, the results of the coarse aggregate strength and abrasion resistance tests demonstrate good performance under traffic loads. Filler testing was carried out to identify the fundamental properties of rice husk ash and sugarcane bagasse ash before their application in the Marshall mix.

Table 1. Physical properties of aggregates.

No.	Test	Unit	Result	Standard Reference	Specification	Remarks
A. Fine Aggregate						
1	Bulk Specific Gravity	-	2.57	SNI 1970: 2008	2.5-2.7	Meets
2	Apparent Specific Gravity	-	2.74	SNI 1970: 2008	-	Meets
3	SSD Specific Gravity	-	2.61	SNI 1970: 2008	2.5-2.7	Meets
4	Water Absorption	%	2.84	SNI 1970: 2008	≤ 3	Meets
B. Coarse Aggregate						
1	Bulk Specific Gravity	-	2.51	SNI 1969: 2016	2.4-2.8	Meets
2	Apparent Specific Gravity	-	2.61	SNI 1969: 2016	-	Meets
3	SSD Specific Gravity	-	2.55	SNI 1969: 2016	2.4-2.8	Meets
4	Water Absorption	%	1.49	SNI 1969: 2016	≤ 3	Meets
5	Asphalt-Aggregate Adhesion	%	97	SNI 2439: 2011	> 95	Meets
6	Aggregate Impact Value		15.79	BS:182 Part 112:1990	≤ 30	Meets
7	Los Angeles Abrasion Test	%	22.73	SNI 2479: 2008	≤ 40	Meets
C Filler						
1	Rice Husk Ash	-	2.26	SNI: 03-4145-1996	2,25-2,27	Meets
2	Bagasse Ash (BA)	-	2,39	SNI: 03-4145-1996	2,25-2,27	Meets

Table 2. Physical properties of asphalt

No	Test	Unit	Result	Standard Reference	Unit
1	Penetration	0.1 mm	68.33	SNI 2456: 2011	≥ 48
2	Softening Point	°C	53	SNI 2434: 2011	≥ 52°C
3	Flash Point	°C	281	SNI 2433-2011	≥232
4	Ductility	cm	> 100	SNI 2432: 2011	≥ 100
5	Asphalt Adhesion to Aggregate	%	96	SNI 2434: 2011	≥ 90
6	Loss on Heating	%	0.19	SNI-06-2440-1991	≤ 0,8%
7	Specific Gravity	-	1.02	SNI 2441: 2011	> 1

As shown in Table 2, based on the laboratory test results of the asphalt material, the result show that all physical parameters—such as penetration (68.3 mm), softening point (53°C), flash point (231°C), ductility (>100 cm), adhesion to aggregate (96%), loss on heating (0.06%), and specific gravity (1.02)—meet the requirements of the Indonesian National Standard (SNI). These results indicate that the asphalt used is of good quality, stable under temperature variations, elastic, and capable of forming a strong bond with aggregates, making it suitable for application in Asphalt Concrete–Binder Course (AC-BC) mixtures.

3.2 AC-BC mixture composition

After the material characteristics were identified, the next step was to determine the composition of the Asphalt Concrete–Binder Course (AC-BC) mixture based on the proportion of aggregate, asphalt, and filler. The composition was arranged to meet the graduation and design requirements for flexible pavement mixtures. The AC-BC mixture composition with the addition of rice husk ash and bagasse ash filler was derived from the Optimum Asphalt Content (OAC) of the standard asphalt mixture. The resulting OAC values are presented in Fig. 3.

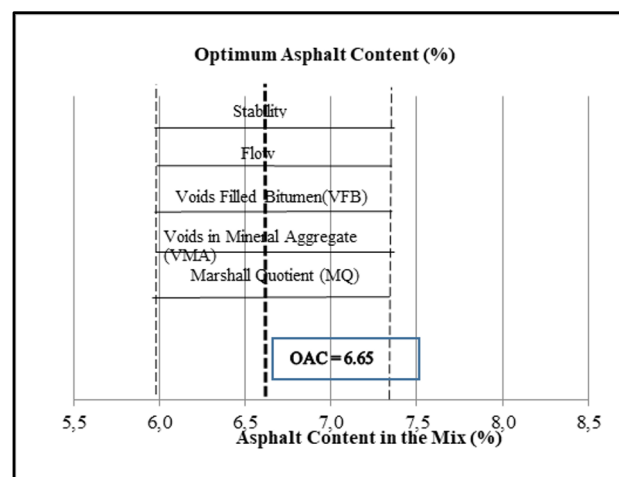


Fig. 3. Optimum asphalt content graph

Figure 3. illustrates the determination graph of the Optimum Asphalt Content (OAC) based on Marshall parameters, such as stability, flow, voids filled with bitumen (VFB), voids in mineral aggregate (VMA), and the Marshall Quotient (MQ). Each horizontal line represents the range of asphalt content percentages in the mixture that meets the technical specifications for each parameter. The graph shows that most of the Marshall parameters fall within the asphalt content range of 6.5%

to 7.3%. The optimum point is indicated by a vertical dashed line at 6.65% asphalt content, which is selected as the OAC. This value is chosen because it offers the best overall performance while meeting all the specifications based on Bina Marga standards. Therefore, an asphalt content of 6.65% is established as the optimum value in this AC-BC mixture design.

From the OAC value shown in Fig. 3, the asphalt mixture composition for the Marshall test with filler

addition can be determined. The detailed composition of the asphalt mixtures is presented in Table 3.

Table 3 shows the AC-BC mix composition with fixed asphalt content (6.65%) and filler variations from 0% to 8%. The total aggregate weight remains constant at 1120.20 g, consisting of coarse, medium, and fine fractions. As fillers increase, fine aggregate decreases to maintain total weight. This data supports the analysis of filler effects on mix performance in the Marshall test.

Table 3. Mix composition of asphalt

Filler %	0	2	4	6	8
Asphalt %	6.65	6.65	6.65	6.65	6.65
Asphalt weight (g)	79.80	79.80	79.80	79.80	79.80
Aggregate weight (gr)	1120.20	1120.20	1120.20	1120.20	1120.20
Coarse Aggregate weight (g)	224.04	224.04	224.04	224.04	224.04
Medium Aggregate weight (g)	582.50	582.50	582.50	582.50	582.50
Fine Aggregate weight (g)	313.66	291.26	268.85	246.45	224.04
Filler weight (g)	0.00	22.40	44.81	67.21	89.62

3.3 Marshall test results with the addition of filler

The Marshall test was conducted to evaluate the performance of the AC-BC mixture based on stability,

flow, and void content parameters. This test serves as the main stage in assessing the effect of using rice husk ash and sugarcane bagasse ash as fillers on the characteristics of the asphalt mixture. The Marshall test results for the mixtures containing rice husk ash and sugarcane bagasse ash are presented in Table 4.

Table 4 Result of the marshall test with the addition of fillers

No.	Filler Content	Specification					
		Stability (Kg)	Flow (mm)	VIM (%)	VMA (%)	VFB (%)	MQ (Kg/mm)
	Unit	≥800	2-4	3.0-5.0	≥14	≥65	≥250
A	Rice Husk Ash						
1	0%	1795.48	3.42	2.34	17.27	82.12	525.51
2	2%	1223.13	2.89	1.92	16.38	87.62	423.72
3	4%	1114.08	0.91	6.52	19.34	72.09	1228.77
4	6%	1009.40	2.95	7.19	20.15	68.63	341.78
5	8%	1039.22	1.72	6.77	19.45	72.19	605.37
B	Bagasse Ash						
1	0%	1341.66	5.63	2.69	17.10	84.86	250.31
2	2%	2527.19	6.20	0.88	15.57	94.36	412.55
3	4%	3453.42	4.83	2.99	17.36	82.97	715.30
4	6%	2739.72	5.03	5.77	19.74	70.82	545.13
5	8%	2783.19	5.20	5.67	19.64	71.30	536.89

The explanation of Table 4 will be described in detail as follows:

3.3.1 Stability

Stability refers to the asphalt mixture's ability to withstand traffic loads without undergoing deformation.

- Rice Husk Ash (RHA): The highest stability was observed at 0% filler content (1795.48 kg) and continued to decrease as the filler content increased, indicating that the addition of RHA reduces the mixture's load-bearing capacity. Thus, as shown in the figure, Bagasse Ash increases plasticity but may lead to bleeding. RHA, on the other hand, tends to result in an overly stiff mixture.

- Bagasse Ash (BA): Shows a significant increase in stability with increasing filler content, peaking at 4% (3453.42 kg), indicating that BA enhances the structural strength of the mixture.

Thus, as seen in Fig. 4, all filler contents meet the specification. The highest value is at 0% (1795.48 kg) and decreases significantly up to 6% (1009.40 kg). Then slightly increases to 8%. This indicates that increasing RHA content generally reduces the mixture's load-bearing capacity. BA all filler contents meet the specification, with the highest value at 4% (3453.42 kg). The increase from 0% to 4% shows a positive effect of BA on structural strength, followed by a slight decline at higher contents, but still above the standard.

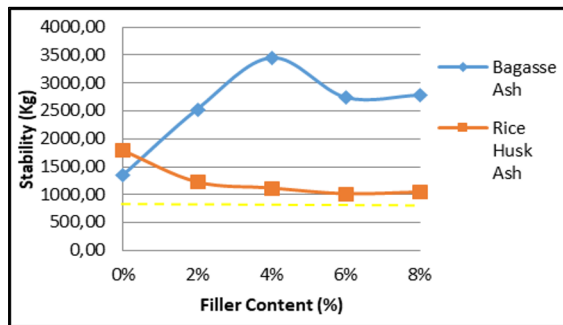


Fig. 4. Stability vs filler content.

3.3.2 Flow

Flow indicates the extent to which the mixture can undergo plastic deformation before failure. The ideal value typically ranges between 2 and 4 mm.

- RHA: Flow values vary, but drop significantly at 4% filler content (0.91 mm), indicating that the mixture is too stiff and prone to cracking.
- BA: Flow values are very high at all filler levels (5.03–6.20 mm), indicating that the mixture is too soft and at risk of permanent deformation (rutting) under heavy traffic.

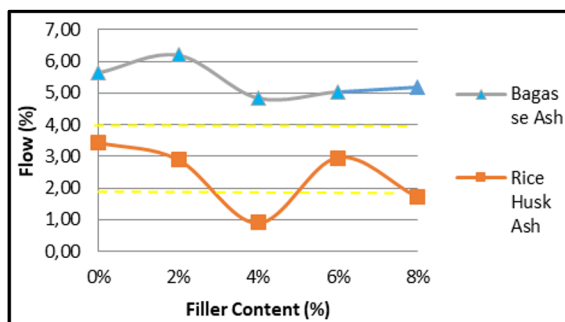


Fig. 5. Flow vs filler content

Thus, as shown in Fig. 5, RHA meets the specification at 0%, 2% and 6%, but is too low at 4% (0.91 mm) and 8% (1.72 mm). Low values indicate a higher risk of brittleness. BA all values exceed the specification range (5.03–6.20 mm), indicating high flexibility but with a greater potential for permanent deformation (rutting) under heavy traffic.

3.3.3 Void in mix (VIM)

VIM indicates the percentage of air voids within the mixture. The ideal value is around 3–5%. Rice Husk Ash (RHA): VIM values increased drastically at 4–8% filler content (>6%), indicating that the mixture became too porous.

Accordingly, as shown in Fig. 6, RHA only at 2% (1.92%) and 3% filler content does the VIM fall within the asphalt content range, while the other filler contents exceed the upper limit, posing a bleeding risk. At 6% and 8%, values are far below the lower limit, indicating excess voids.

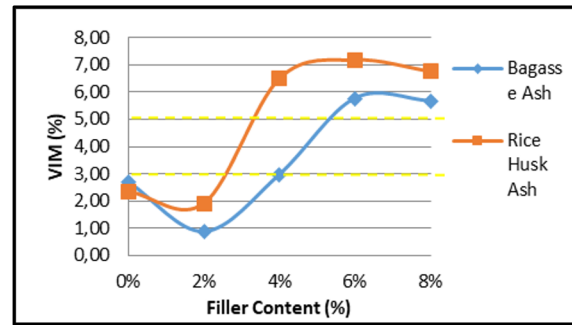


Fig. 6 VIM vs filler content

3.3.4 Void in mineral aggregate (VMA)

RHA and BA: Both fillers show VMA values that meet the specifications, although BA has slightly lower values at the beginning (15.57% at 20%). Interpretation: Both fillers maintain the inter-aggregate volume within a safe range.

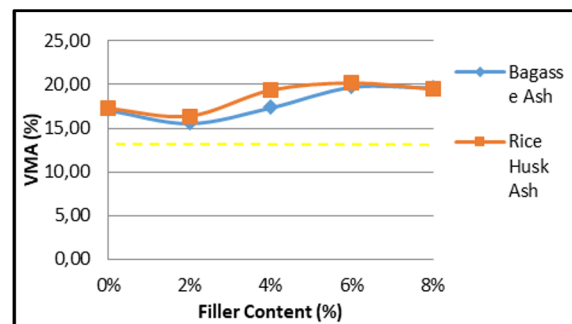


Fig. 7 VMA vs filler content

Thus, as shown in Fig. 7, all values of RHA and BA meet the specification. RHA generally produces slightly higher VMA than BA, meaning there is sufficient space for asphalt film coating around the aggregate particles.

3.3.5 Void filled with bitumen (VFB)

VFB is the percentage of VMA filled with asphalt.

- RHA: VFB values decrease with increasing filler content but remain within specification limits.
- BA: VFB values tend to be high (>84%), indicating a potential for bleeding, especially at lower filler contents.

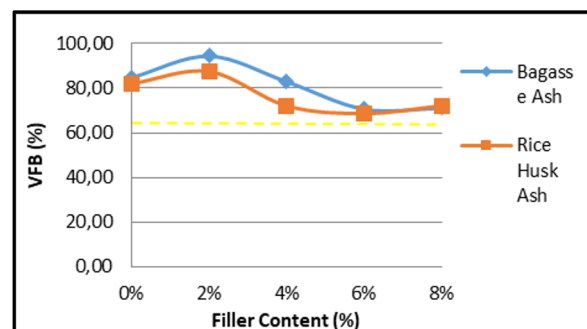


Fig. 8. VFB vs filler content.

Thus, as seen in Fig. 8, both BA and RHA mixtures met the VFB specification, as all values exceeded the

lower limit of 65%, indicating that the voids were sufficiently filled with asphalt to ensure proper coating and binding.

3.3.6 Marshall quotient

Rice Husk Ash (RHA) significantly increases the stiffness of the mixture at 4% content, but this can lead to cracking due to excessive rigidity. In contrast, Bagasse Ash (BA) provides more stable MQ values, resulting in a well-balanced mixture with both strength and flexibility, making it more resistant to deformation without being prone to cracking.

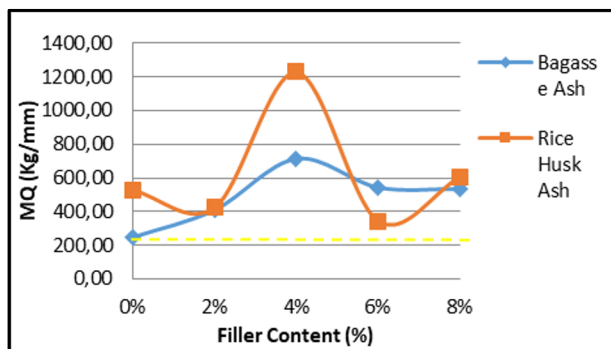


Fig. 9. MQ vs filler content

Thus, as shown in Fig. 9, RHA values meet the specification. The highest MQ is at 4% (1228.70 Kg/mm), indicating high stiffness, but excessive stiffness can reduce flexibility. BA all values meet the specification, showing an optimal balance between strength and deformation resistance.

The trends observed in stability and flow data suggest that both types of ash exhibit optimal performance at moderate filler content (4-6%) with diminishing results beyond this range. This can be attributed to excessive filler interfering with the optimal bitumen aggregate matrix, potentially leading to increased stiffness and brittleness. The higher Marshall Quotient (MQ) observed in BA mixtures indicates better load-bearing capacity and resistance to deformation, likely due to the angular and porous structure of BA particles that enhances interlock and binder absorption.

3.4 Comparative analysis

The comparative analysis of the Marshall test results demonstrates clear differences in performance between Rice Husk Ash (RHA) and sugarcane Bagasse Ash (BA) as fillers in AC-BC mixtures. In terms of stability, RHA exhibited a decreasing trend with increasing filler content, indicating reduced load-bearing capacity due to increased mixture stiffness. In contrast, BA significantly improved stability, with the highest value observed at 4% filler content, suggesting its superior ability to enhance structural strength. Regarding flow, RHA resulted in values below the standard range, indicating a stiff and potentially brittle mix. At the same time, BA produced flow values above the acceptable range, suggesting

excessive softness and a risk of deformation under traffic loads.

For voids in mix (VIM), RHA caused higher porosity at greater filler contents, whereas BA maintained VIM values within the ideal range at 4–8% filler levels. Both fillers yielded acceptable voids in mineral aggregate (VMA) values, though BA showed slightly more consistent results. In terms of voids filled with bitumen (VFB), RHA provided more balanced values across filler variations, while BA tended to retain excessive asphalt at low contents, risking bleeding. However, BA demonstrated more stable Marshall Quotient (MQ) values, reflecting a better balance between strength and flexibility compared to RHA, which showed excessively high stiffness at 4% filler content.

3.5 Limitations and future work

Based on the overall findings, sugarcane bagasse ash (BA) is identified as the more effective filler material in AC-BC mixtures. Its ability to significantly increase mixture stability, maintain balanced mechanical characteristics, and provide consistent performance across several Marshall parameters makes it more suitable than RHA. Additionally, as an agricultural waste product, BA represents a sustainable and environmentally friendly alternative to conventional fillers. The optimum performance of BA was achieved at a 4% filler content, making it a highly recommended choice for enhancing the durability and strength of asphalt concrete wearing courses.

Higher stability and MQ values in BA mixtures could be attributed to better filler bitumen adhesion, supported by previous SEM observation on BA morphology, indicating porous and angular particles that enhance interlocking.

Although this study demonstrates that both RHA and BA can serve as alternative fillers, their performance is influenced by their distinct chemical composition. The superior performance of BA and RHA is likely attributed to its higher pozzolanic activity and better particle morphology. RHA, rich in silica, may lead to excessive stiffness and brittleness at higher contents due to limited calcium oxide or other binding components. Future research should include a detailed chemical and microstructural analysis to better explain the mechanism behind the differences in performance. This difference in performance is attributed to the higher reactivity and pozzolanic index of Bagasse Ash due to its higher CaO and amorphous silica content. Whereas RHA, despite its high SiO₂ content, has a denser structure and lower pozzolanic activity.

4 Conclusion

This study aimed to identify the impact of using Agricultural Waste-Rice Husk Ash (RHA) and Bagasse Ash (BA) as fillers in Asphalt Concrete (AC-BC) mixtures. The findings clearly demonstrate that both fillers affect the mechanical performance parameters defined by the Marshall test.

1. Stability increased significantly with BA, peaking at 3453,42 Kg at 4% filler content, indicating enhanced load-bearing capacity. In contrast, RHA showed the highest stability (1795.48 kg) at 0% filler and decreased as filler content increased, suggesting a reduction in load capacity at higher dosages.
2. Flow values revealed that excessive RHA content ($\geq 4\%$) reduced flexibility to 0.91 mm, causing brittleness, while BA produced higher than standard values (5,03-6,20 mm), indicating greater flexibility but potential permanent deformation.
3. Void in Mix (VIM) for RHA exceeded the ideal 3-5% range at higher filler contents, whereas BA maintained VIM within acceptable limits at 4-8%, though too low (0.88%) at 2% filler.
4. Void in Mineral Aggregate (VMA) remained within standards for both fillers, ensuring adequate binder coverage, with RHA generally giving slightly higher values. Exceeded 84% at lower contents, increasing bleeding risk
5. Void-Filled with Bitumen (VFB) were more balanced in RHA mixtures, while BA achieved its peak (525.52 kg/mm) at 0% filler but dropped sharply with higher dosages.

Overall, BA at 4% filler content is most effective for improving structural performance, offering higher stability and balanced flexibility, while RHA is better applied at lower contents to avoid excessive stiffness. Both materials meet Indonesia National Standards and support sustainable pavement construction by recycling agricultural waste into high-value fillers.

Future work should explore long-term durability and binder-filler interaction mechanisms through microstructural analysis to optimize their application in flexible pavement.

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